Seed Dispersal Derby

Engineer a seed that escapes its parent tree

At a glance: Build a seed that escapes the parent tree without engines or throwing. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: adaptation, drag, lift, mass, surface area, dispersal strategy (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- paper and cardstock: shared, about 6 to 8 sheets per team (4 teams)
- paper clips: a few per team for mass, one small box covers the camp
- masking tape: shared, 1 roll plus 1 spare
- string: shared, to mark the standard drop height
- fan or marked drop lane: 1 shared test zone on a low setting
- tape measure: 1 shared, to measure distance traveled
- timers: 1 per team (4), plus 1 spare or staff phones

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up one station per team with the materials above, sorted and ready.

02 Stage the scoring sheet and any reference values before testing begins.

03 No PPE is needed: this is a dry paper build tested at a low-power fan or a marked drop lane, with no soil, water, or chemicals. Instead, set the fan to its lowest setting, mark the launch lane and target away from people, and clear loose paper clips.

04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Build a seed that escapes the parent tree without engines or throwing." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Test and record 0:44 - 0:56

Teams run the standard test (drop from the marked height or release in the low fan lane), time three trials and average the hang time, measure the distance with the shared tape, and log this baseline before changing anything.

Phase 6 Redesign 0:56 - 1:08

Each team changes exactly one variable (wing length, fold angle, OR added mass), predicts the effect, and re-tests with the same three-trial standard test to compare hang time and distance against baseline.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Collect paper scraps and loose paper clips and return the fan, timers, and tape measure to the bin; there is no PPE and nothing to wash.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: No measurable result on the first test. **Fix:** Check the setup against the steps, confirm the standard test was followed, and log the baseline anyway.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to drag, lift, and hang time.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which change improved hang time without losing distance?
- Why does spinning slow the fall?
- Which tree's strategy did your final seed copy?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Distance traveled	35
Hang time	25
Target landing	15
Biomimicry explanation	15
Redesign improvement	10
Total	**100**

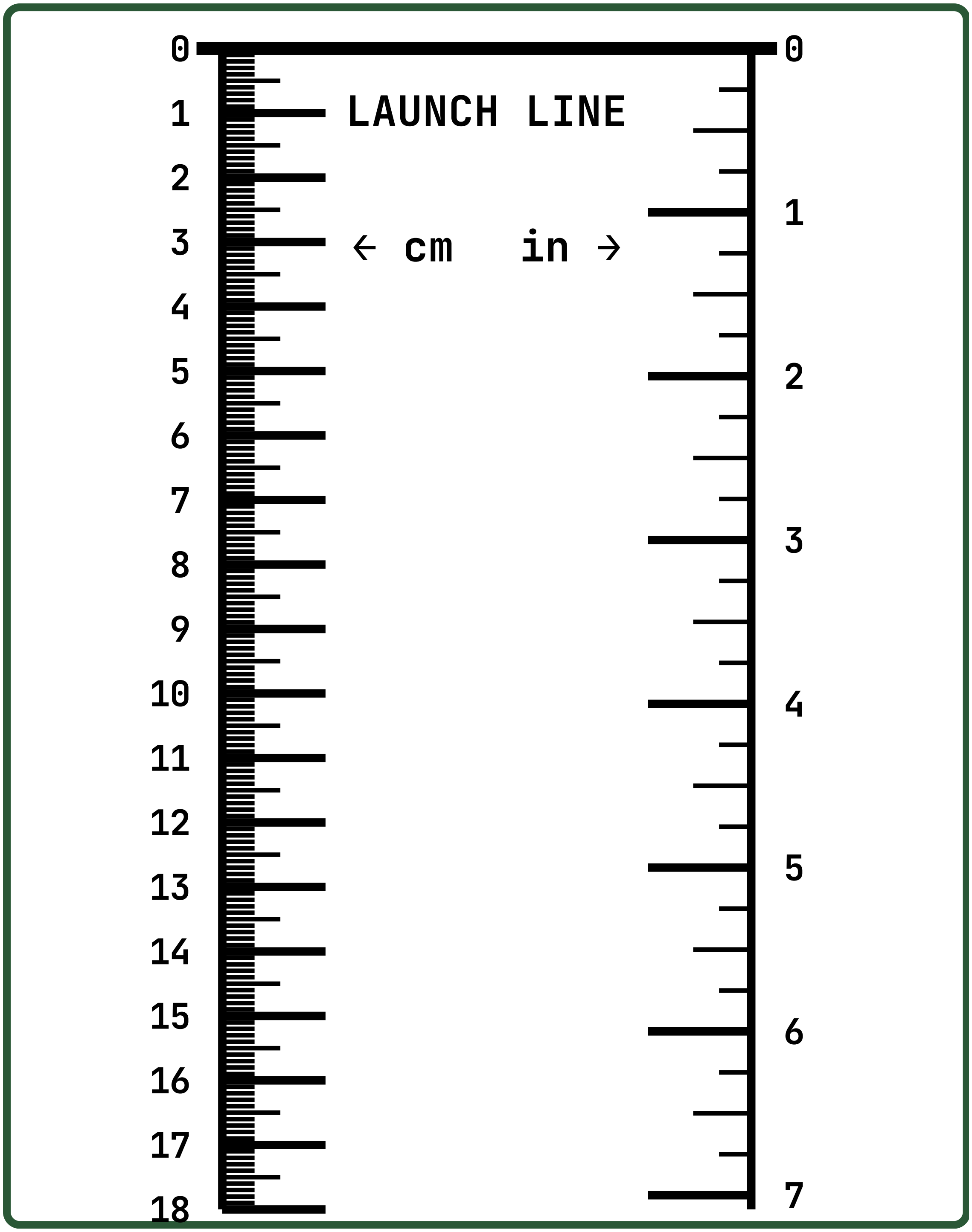
CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES Science Buddies Seed Dispersal; Project Learning Tree

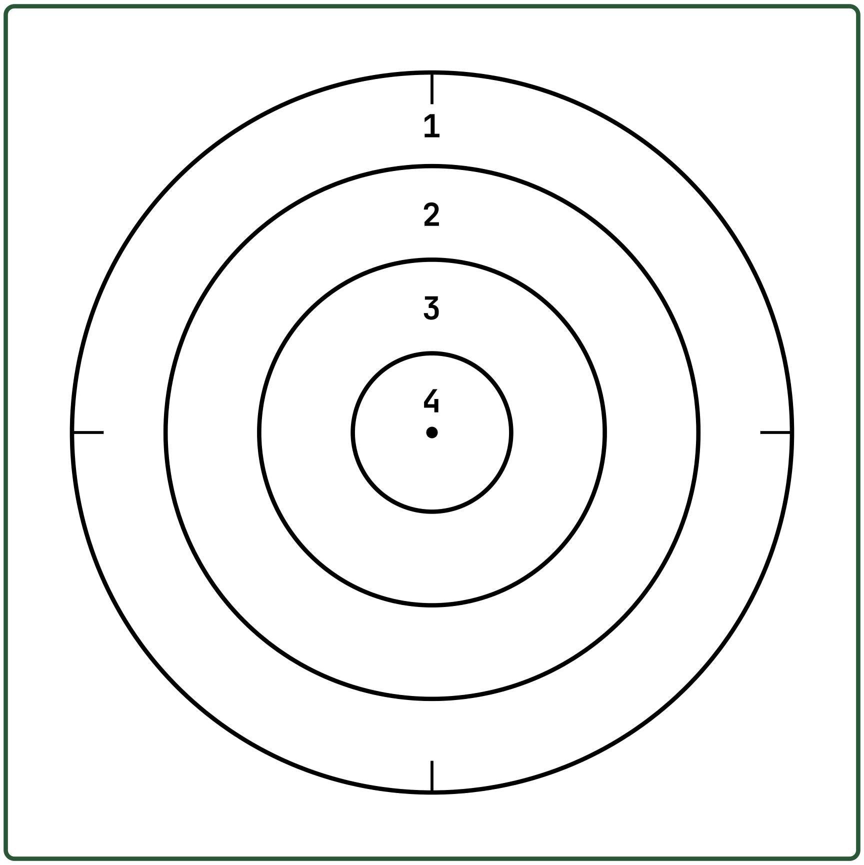
Print at 100% (not "fit to page") on cardstock so the centimeter ruler stays true to size. Tape the strip flat down the lane with the **0** end at the launch line; for a longer lane print several strips and tape them head to tail.

Drop-lane distance strip



Landing-zone target

Lay the target flat where staff pick down the lane and line the crosshairs up with the lane center. A higher ring number is a better landing.



Ring 1
Outer ring. Your seed touched the target.

Ring 2
Good aim. Getting closer.

Ring 3
Close to the center.

Ring 4
Bullseye, right on the dot.